



Facility of engineering and technology

Computer Science Department

COMP4381(Cloud computing and data security)

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Student ID : 1221971

Assignment: Cloud computing Virtualization

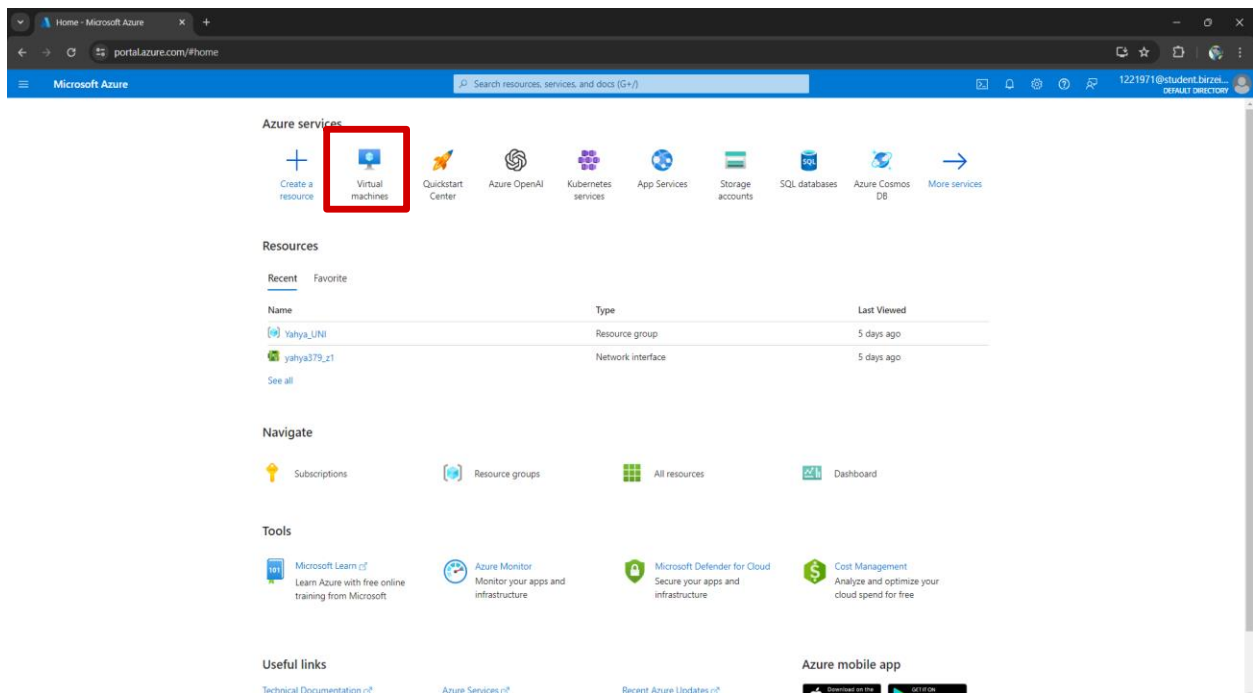
The Scenario: You are a baker and want to showcase your delicious treats online. Build a simple website with your bakery information and contact details. Build the website on a Virtual Machine (VM) in the Azure portal.

Steps :

1- Creating the Virtual Machine (VM):

Cloud Computing Service provider used : Microsoft Azure

After creating an account in Microsoft Azure using the University Email (.edu) we will get this page to start with :



From the Azure Services bar shown above we choose the first option which is Virtual Machine (in the red Square)

After choosing (VM) option this page will appear to start give the VM that we need to create specific features as needed :

Microsoft Azure

Home > Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * Create new group

Create new

Instance details

Virtual machine name *

Region * (Africa) South Africa North

Availability options Availability zone

Availability zone * Zones 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type Trusted launch virtual machines

Configure security features

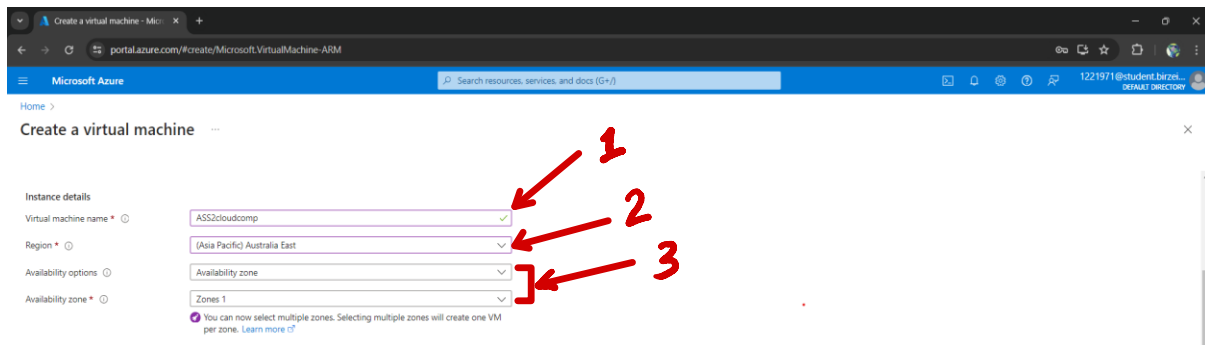
< Previous Next : Disks > Review + create

Give feedback

As shown in the screen shot the subscription used is Azure for Students (free tier).

Also we created a new Resource group named (Yahya_BZU2) as shown from the button (Create new) in red square.

Now we need to complete choosing the features for the VM :



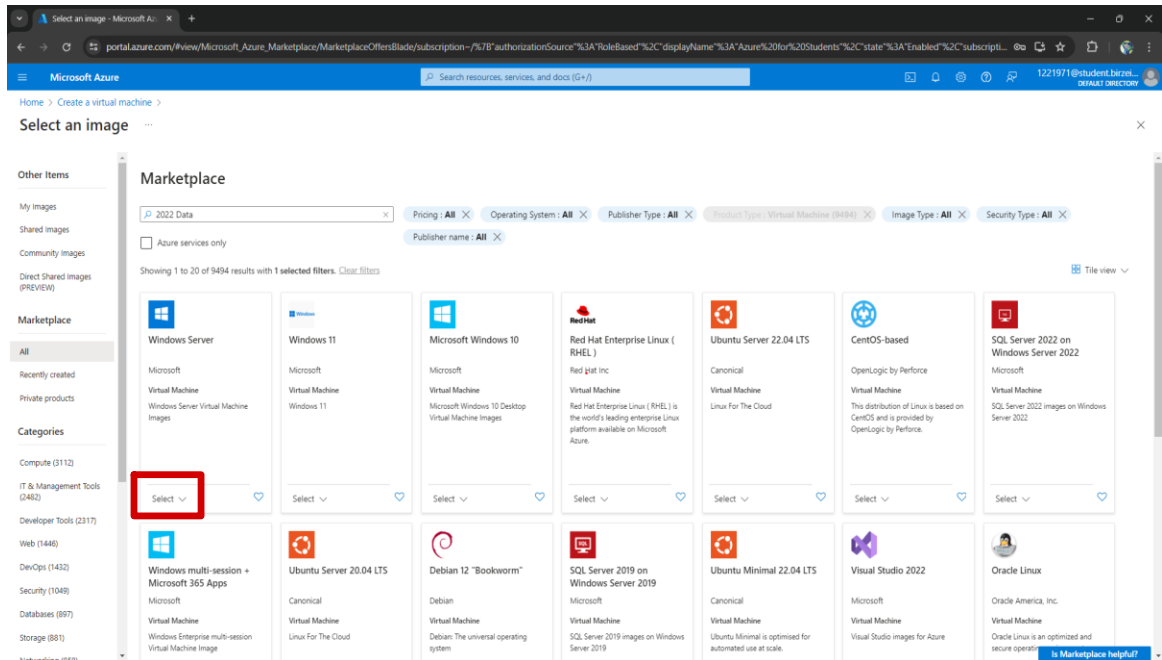
Description :

- 1- Just a name for the VM in the service provider to identify it when you have more VMs.
- 2- Choosing the region where your VM will be (Server), you need to choose one that is close to your current location and has a good connection to your current location.
- 3- More advanced settings to the availability in the chosen region.

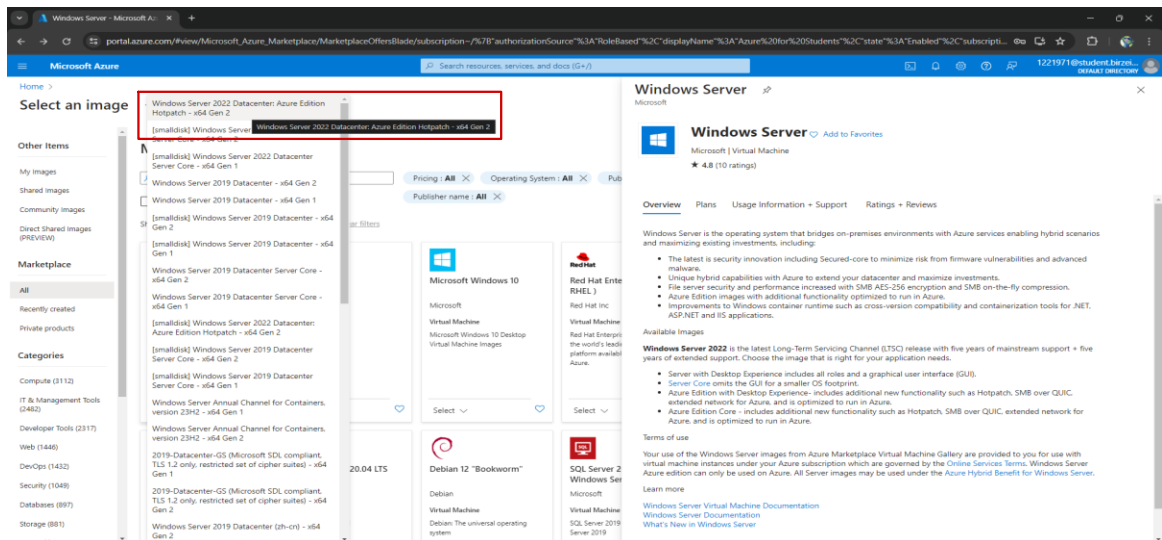
Now we need to choose the image(Operating System) that we want it in our VM :



Click (see all images) to search for the OS you need or see all of them , after that this page will appear :

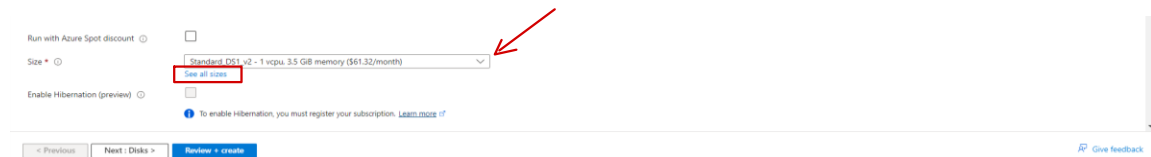


For the OS we need , click (select) as shown above(Red square) then , this page will appear :



This shows all the versions of the OS chosen (we need the first one as shown in the **red box**).

After choosing the OS we need , we have to configure the size of the VM , which means how much you need computing power and computer resources from the cloud service provider :



Click (See all sizes) , to see all the sizes available , after click this page will appear :

VM Size	Type	vCPUs	RAM (GiB)	Data disks	Max IOPS	Local storage (GiB)	Premium disk	Cost/month
B2ms	General purpose	2	8	4	1920	16 (SCS)	Supported	\$83.22
DS2_v2	General purpose	2	7	8	6400	14 (SCS)	Supported	\$205.13
B4ms	General purpose	4	16	8	2880	32 (SCS)	Supported	\$165.71
D4L_v3	General purpose	4	16	8	6400	32 (SCS)	Supported	\$316.82
DS3_v2	General purpose	4	14	16	12800	28 (SCS)	Supported	\$410.26
D-Series v4 The 4th generation D family sizes for your general purpose needs								
B-Series Ideal for workloads that do not need continuous full CPU performance								
B2s	General purpose	2	4	4	1280	8 (SCS)	Supported	\$44.38
B1s	General purpose	1	1	2	320	4 (SCS)	Supported	\$12.56
B2ms	General purpose	2	8	4	1920	16 (SCS)	Supported	\$83.22
B4ms	General purpose	4	16	8	2880	32 (SCS)	Supported	\$165.71
B1ms	General purpose	1	2	2	640	4 (SCS)	Supported	\$22.19
E-Series v4 The 4th generation E family sizes for your high memory needs								
F-Series v2 Up to 2X performance boost for vector processing workloads								
D-Series v3 The 3rd generation D family sizes for your general purpose needs								
E-Series v3 The 3rd generation E family sizes for your high memory needs								
D-Series v2 The 2nd generation D family sizes for your general purpose needs								

The size we choose is (Basic B1s) the suggested size for our purpose ,and the price of the size you choose means the expected cost in a month , but if you don't use the VM you don't pay anything because in Cloud computing you pay as much as you use .

Now we need to put a username and password to the actual VM we are creating and we use them to enter to the VM and access it remotely ,as shown :

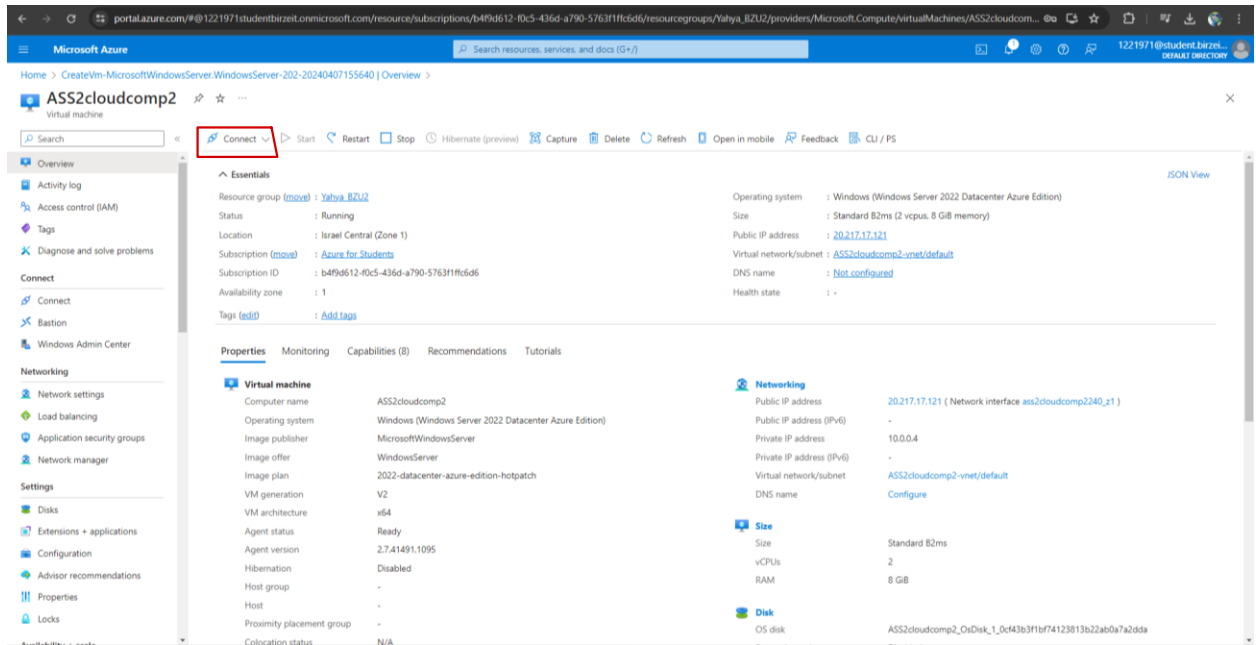
Microsoft Azure portal showing the 'Create a virtual machine' configuration page. The 'Administrator account' section is highlighted, showing the 'Username' field set to '1221971' and the 'Password' field set to a masked password. The 'Confirm password' field is also masked. The 'Review + create' button is highlighted at the bottom.

Username : 1221971

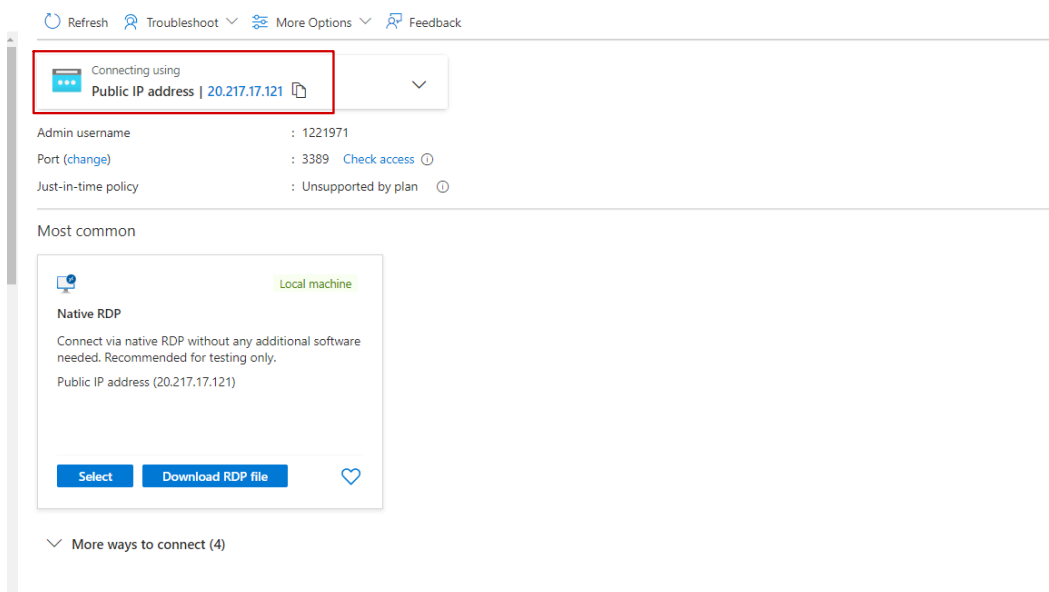
Password : 1221971@YahyaAburayyan

2- Connect to your (VM) :

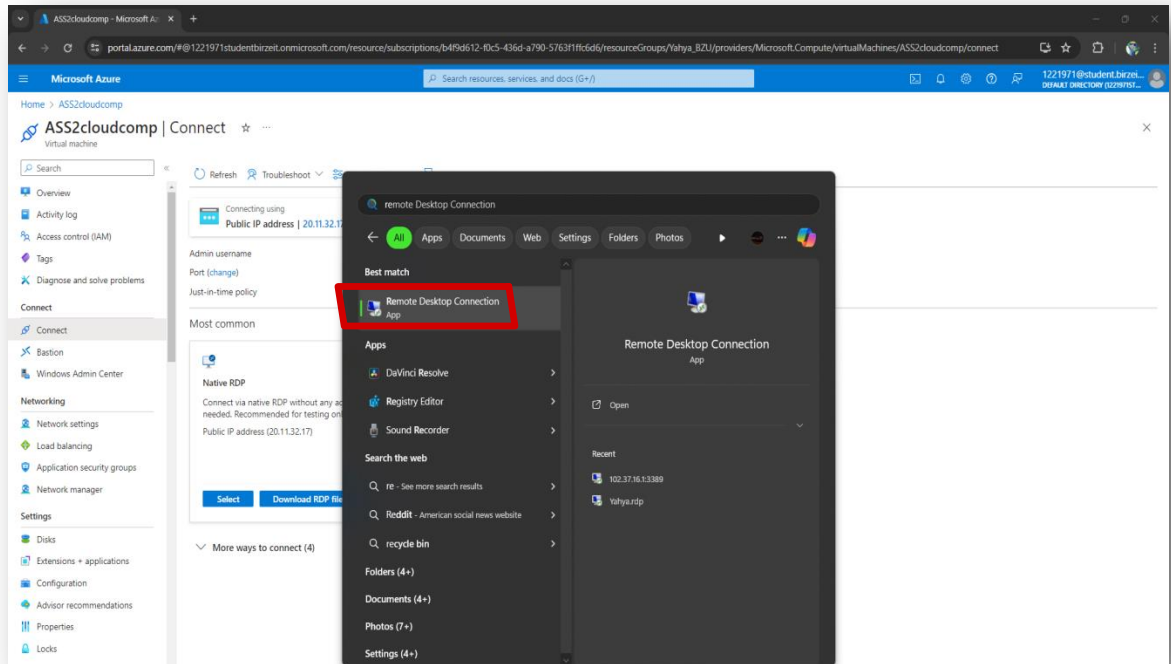
Now we click (Review + create) button and finish creating our VM , then this summary page will appear :



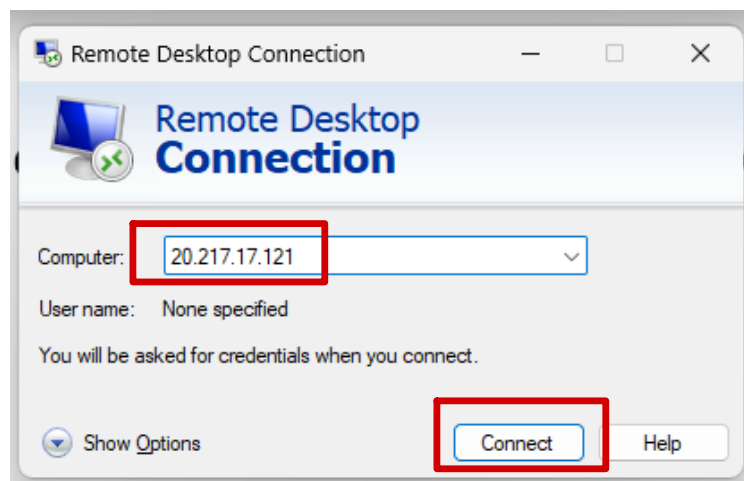
After this page appear our VM is done , we need to connect to it to control it remotely , click on (Connect) as shown ,after that this page will appear :



Now we need to copy the IP address to connect to the server has our VM in the cloud service provider , then search for windows app called (Remote Desktop Connection) to use the VM :



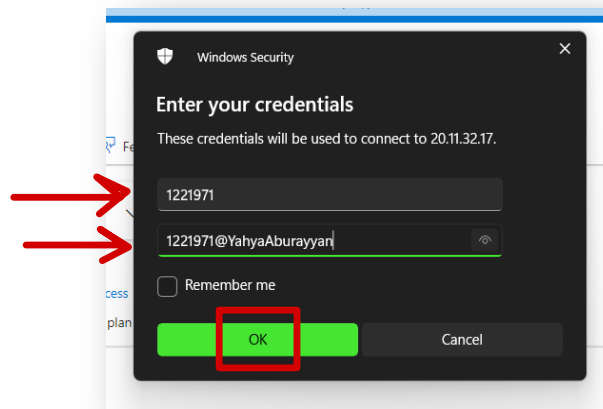
After running the tool this will be the first window on the tool :



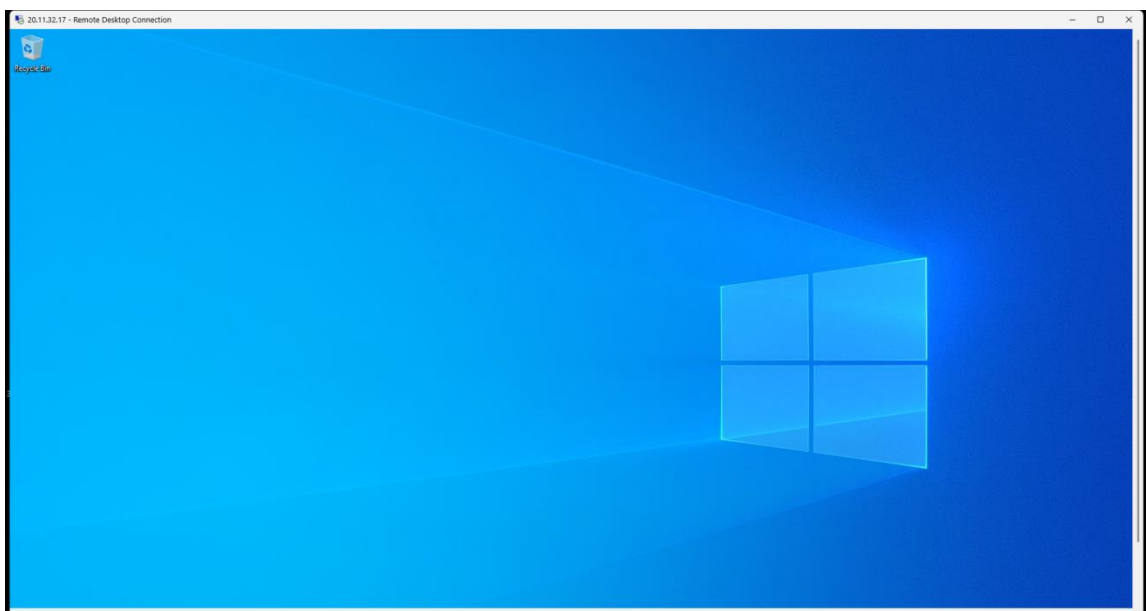
Now we enter the copied IP address and fill it in the (Computer) as showed above , and click (Connect) we connect using RDP.

IP Address : 20.217.17.121

Now in this page we need to enter to the VM using the username and password we used in creating the VM above as follows :

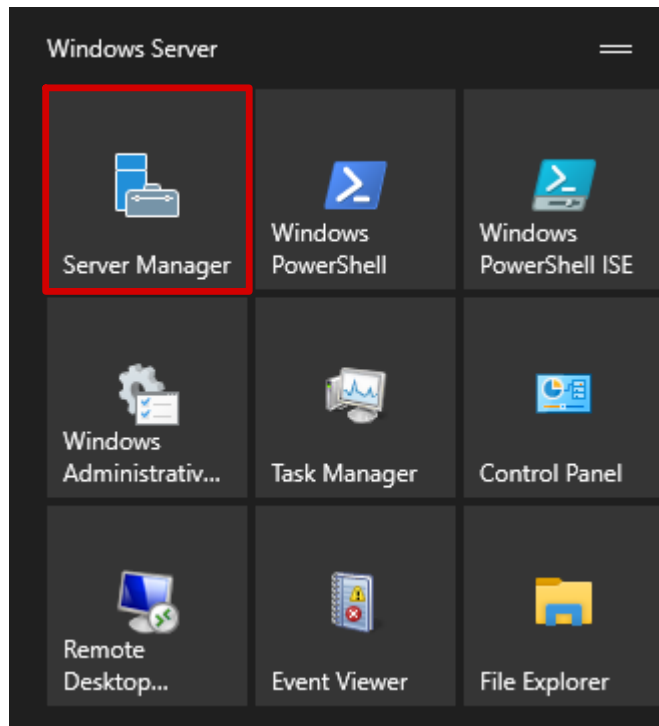


After click (OK) we will connect remotely to the VM that we created and be able to use it like a normal windows system :

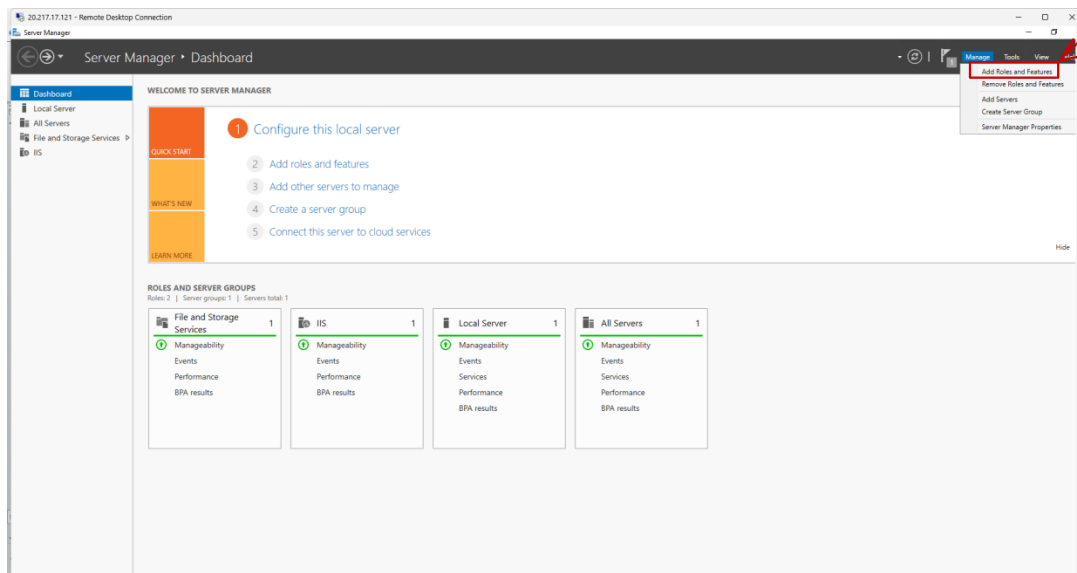


3- Install Web Server (Web Server Configuration):

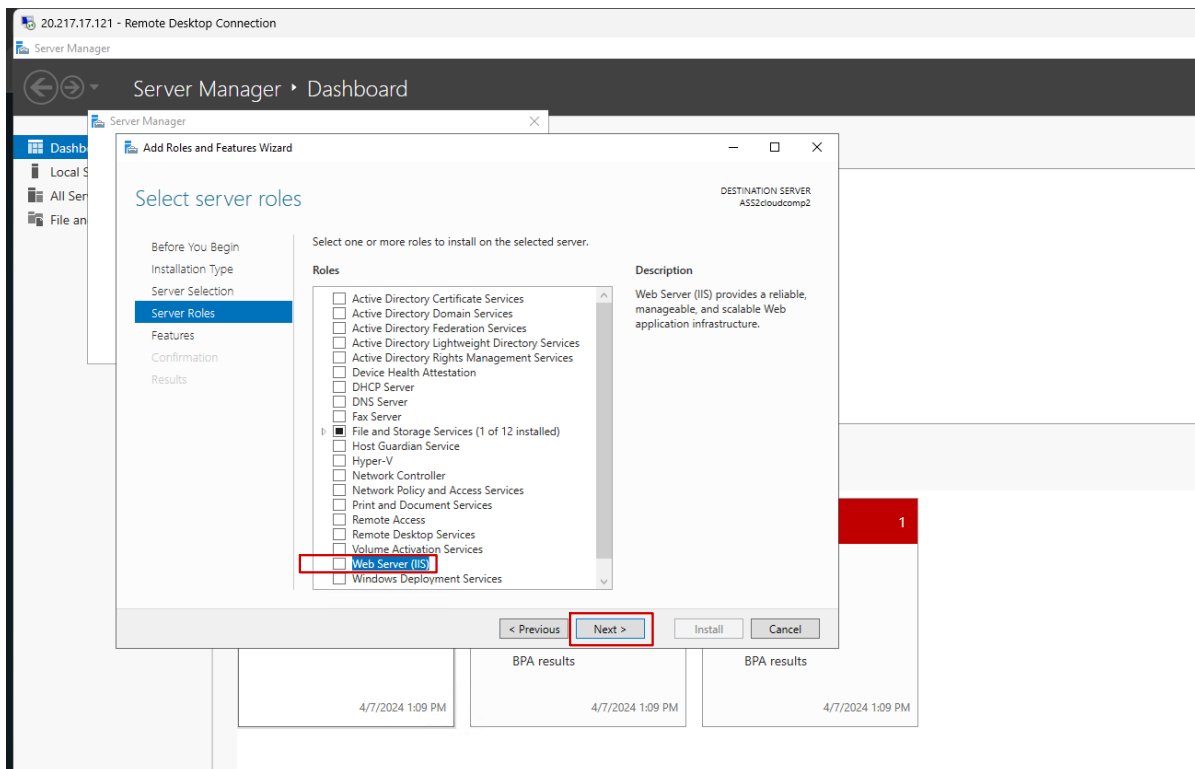
We need to open the (Windows server 2022 datacenter) Server manager :



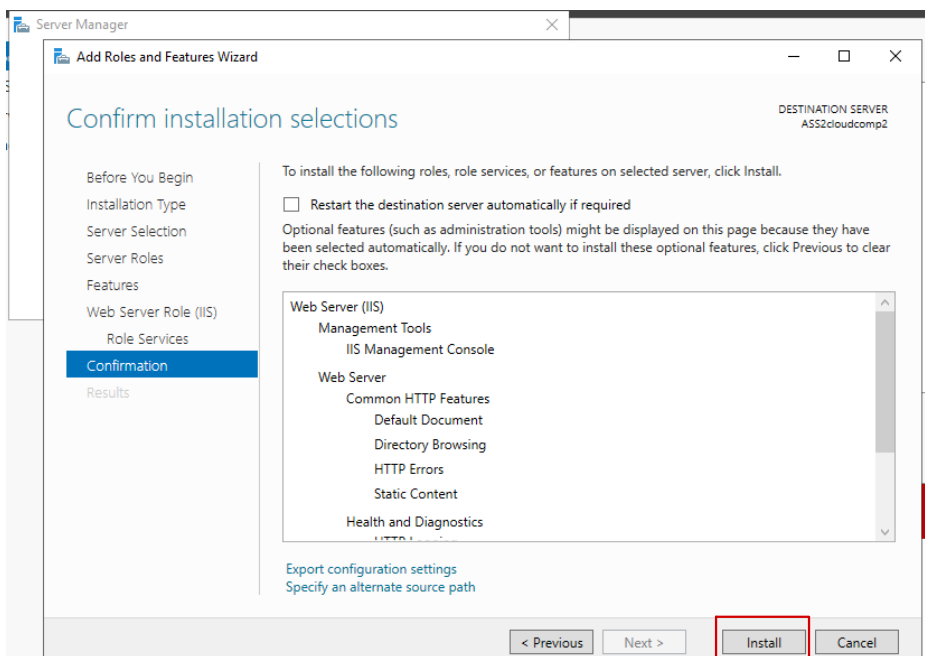
After open it we need to install the (IIS) , so choose manage then (Add Roles and Features) :



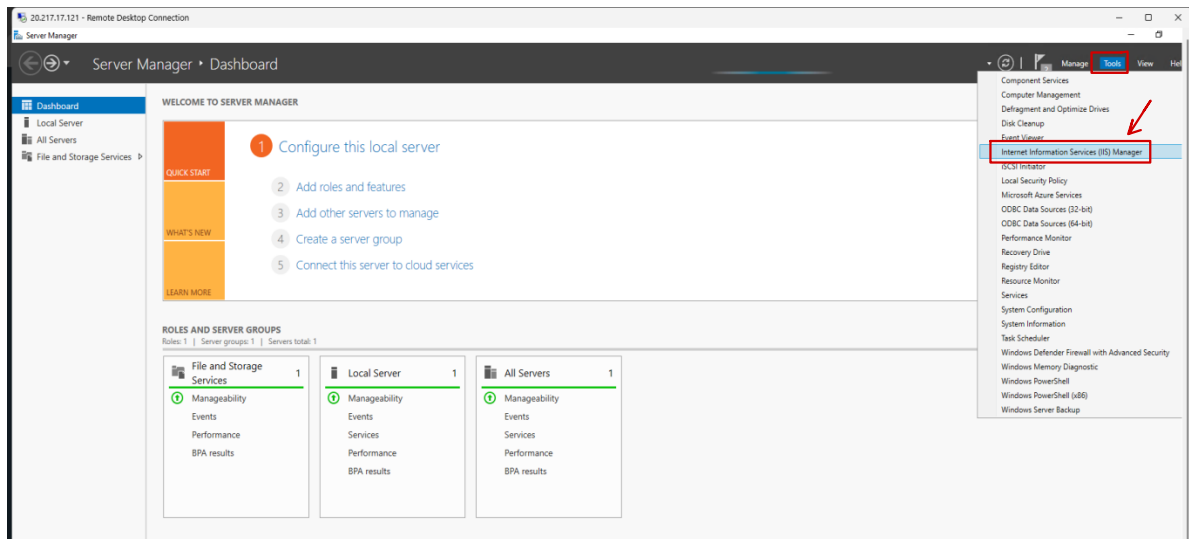
Now we choose (Web Server (IIS)) to be installed :



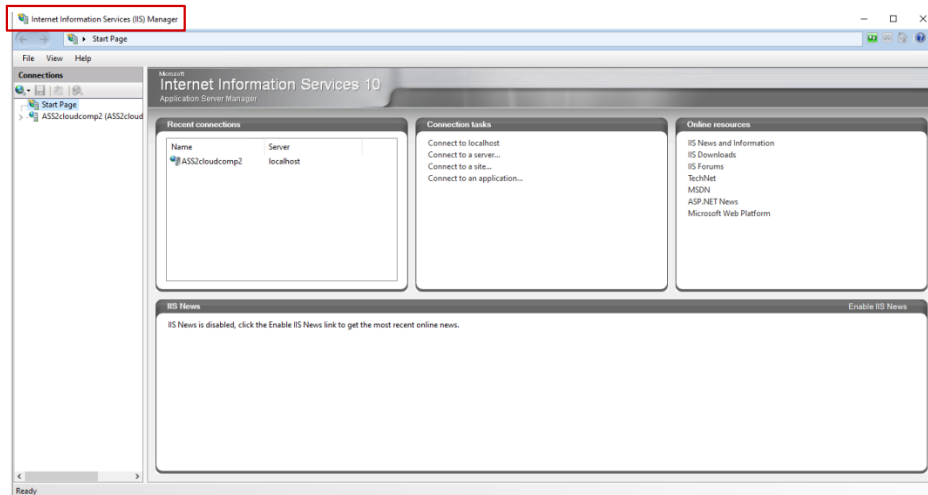
Keep click next till this page and click install :



After finishing the install process , from the home page , from tools select (IIS tool) as follows :



Now this is the main page for the (IIS Manger) :



4- Create Simple Web Page :

After installing any code editor , we will write a simple HTML page about the scenario that our assignment is about , and save the file in our VM in this format :

Index.html

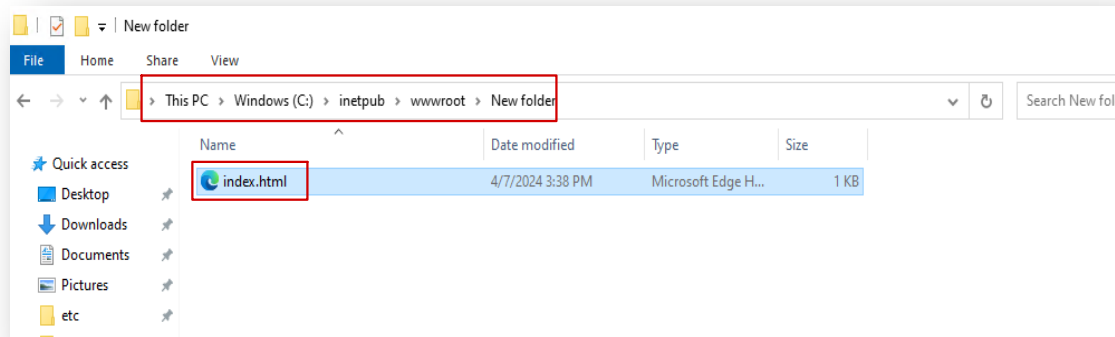
By using VS Code, My html code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>BZUBacary</title>
</head>
<body>
  <h1>Welcome to BZUBacary!</h1>
  <p>This is a simple webpage for BZUBacary.</p>

  <h2>Contact Information</h2>
  <p>Contact Number: +972589989</p>

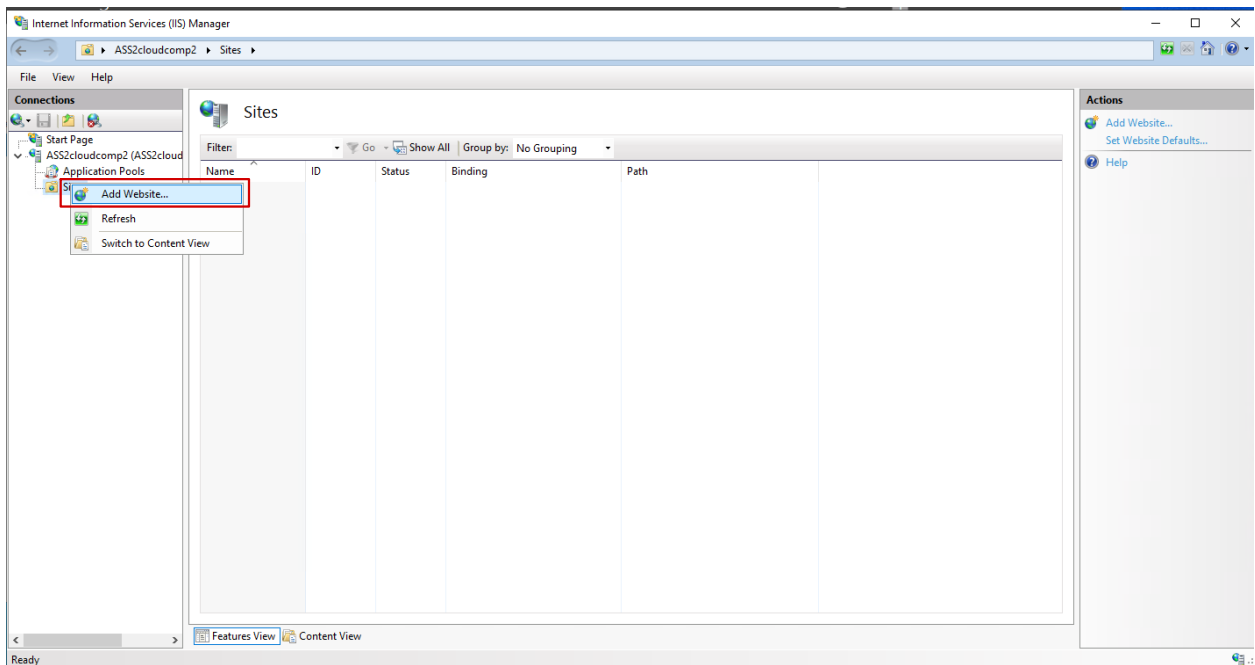
  <h2>Social Media</h2>
  <ul>
    <li><a
href="https://www.facebook.com/bacary">Facebook</a></li>
    <li><a
href="https://twitter.com/bacary">Twitter</a></li>
    <li><a
href="https://www.instagram.com/bacary">Instagram</a></li>
  </ul>
</body>
</html>
```

Saving the index.html file location (path) in the VM :



5- Configure the web server:

Now we need to configure the IIS to host our HTML page, from the IIS manager select (Add Website) as showed below :



Now this page will appear to give our website name, host name and others fill it and click (OK) as showed :

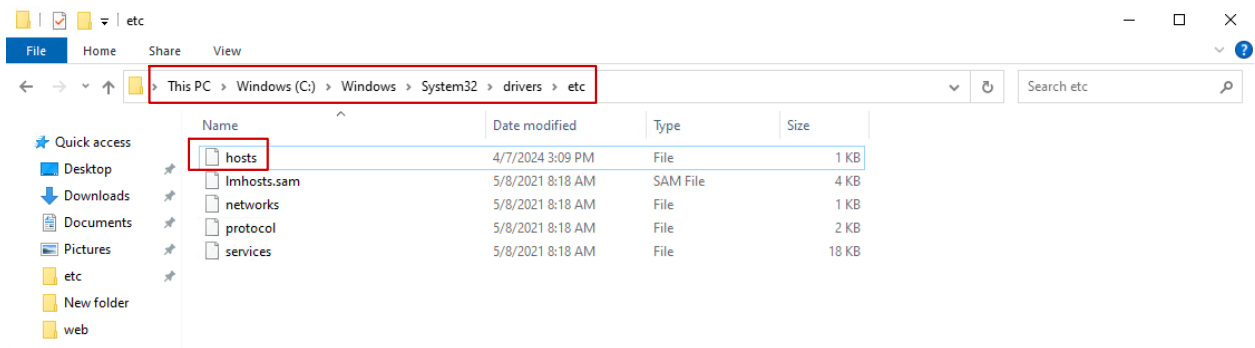
The screenshot shows the 'Add Website' dialog box with the following fields and values:

- Site name:** BZUBacary.com
- Application pool:** BZUBacary.com
- Content Directory:**
 - Physical path:** C:\inetpub\wwwroot\New folder
 - Pass-through authentication:** (disabled)
 - Buttons:** Connect as..., Test Settings...
- Binding:**
 - Type:** http
 - IP address:** All Unassigned
 - Port:** 80
 - Host name:** BZUBacary.com
 - Example:** www.contoso.com or marketing.contoso.com
- Start Website immediately:** ☒
- Buttons:** OK, Cancel

Red arrows indicate the following fields to be filled:

- Site name
- Physical path
- Type
- Host name
- OK button

To make this web page run we need to add it's public IP Address to a specific file in the VM OS the file name is : (hosts) as this path show :



Now open the hosts file and append this IP and save as follows :

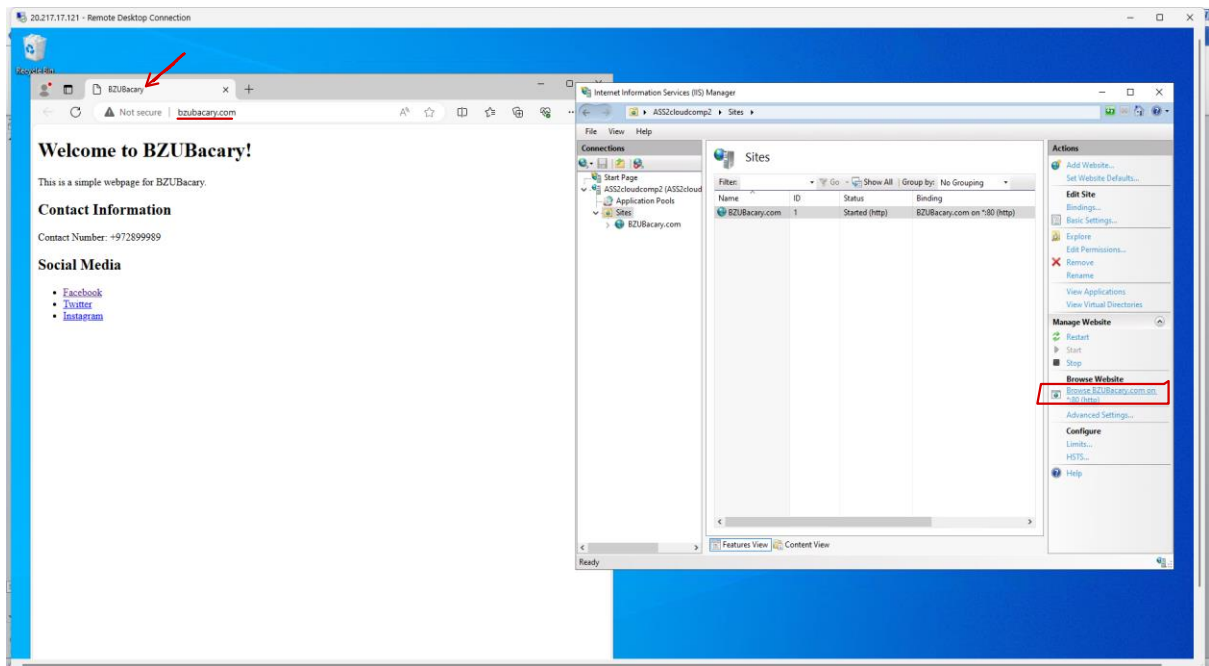
The screenshot shows a Notepad window titled '*hosts - Notepad' with the following text:

```
File Edit Format View Help
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com           # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1        localhost
#       ::1              localhost
127.0.0.1 BZUBacary.com
```

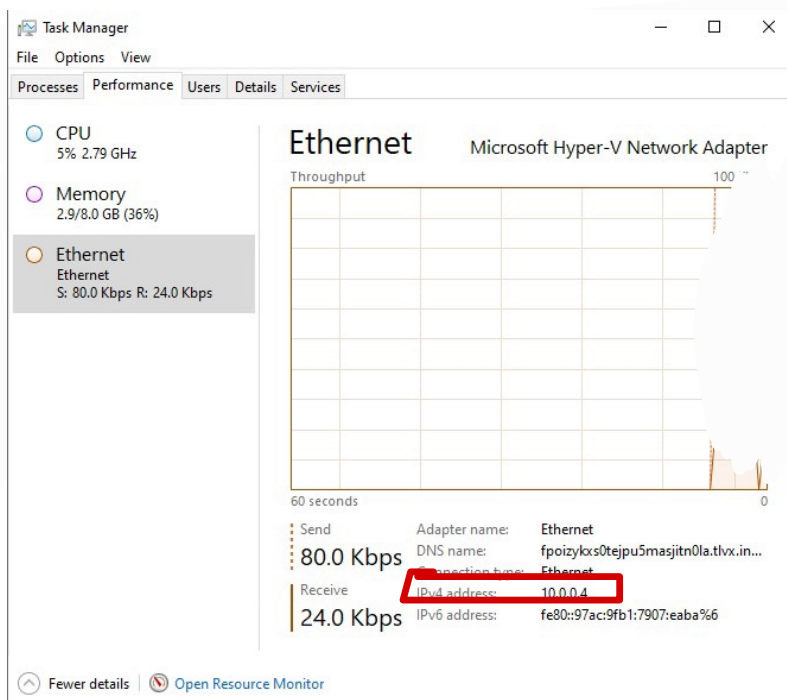
The status bar at the bottom indicates 'Ln 22, Col 24', '100%', 'Windows (CRLF)', and 'UTF-8'.

Now back in the IIS manager if we click Browse Website as shown our website will open as this :



Now our bakery website is hosted on the web server .

6- Test our Website :



20.217.17.121 - Remote Desktop Connection

BZUBacary x +

< > Not secure | 10.0.0.4

Welcome to BZUBacary!

This is a simple webpage for BZUBacary.

Contact Information

Contact Number: +972899989

Social Media

- [Facebook](#)
- [Twitter](#)
- [Instagram](#)